

```
Ubuntu
amit07@DESKTOP-AMIT07:/mnt/c/Users/user/Desktop/ppwc_Project$ gcc -Wall -Wextra
-std=c99 -o cat cat.c
amit07@DESKTOP-6698H4L:/mnt/c/Users/user/Desktop/ppwc_Project$
```

**Figure 1: Compilation of cat.c program using GCC compiler**

```
amit07@DESKTOP-AMIT07:~/ppwc$ echo "Line 1: Hello World" > test1.txt
amit07@DESKTOP-AMIT07:~/ppwc$ echo "Line 2: This is a test file" >> test1.txt
amit07@DESKTOP-AMIT07:~/ppwc$ echo "Line 3: End of file" >> test1.txt
amit07@DESKTOP-AMIT07:~/ppwc$ echo "Second file content" > test2.txt
amit07@DESKTOP-AMIT07:~/ppwc$ echo "More data in file 2" >> test2.txt
amit07@DESKTOP-AMIT07:~/ppwc$
```

**Figure 2: Create test files**

```
amit07@DESKTOP-AMIT07:~/ppwc$ ./cat
This is cat commandThis is cat commandamit07@DESKTOP-AMIT07:~/ppwc$
```

**Figure 3: Test Case 1 - Reading from standard input (stdin)**

```
This is cat commandThis is cat commandamit07@DESKTOP-AMIT07:~/ppwc$ ./cat test1.txt
Line 1: Hello World
Line 2: This is a test file
Line 3: End of file
```

**Figure 4: Test Case 2 - Displaying contents of a single file**

```
amit07@DESKTOP-AMIT07:~/ppwc$ ./cat ">" "output.txt"
This is also cat commandamit07@DESKTOP-AMIT07:~/ppwc$ cat output.txt
This is also cat commandamit07@DESKTOP-AMIT07:~/ppwc$
```

**Figure 5: Test Case 3 - Output redirection with overwrite mode**

```
This is also cat commandamit07@DESKTOP-AMIT07:~/ppwc$ ./cat ">>" "output.txt"
Why is this a cat command broamit07@DESKTOP-AMIT07:~/ppwc$ cat output.txt
This is also cat commandWhy is this a cat command broamit07@DESKTOP-AMIT07:~/ppwc$
```

**Figure 6: Test Case 4 - Output redirection with append mode**

```
amit07@DESKTOP-AMIT07:~/ppwc$ ./cat test1.txt test2.txt
Line 1: Hello World
Line 2: This is a test file
Line 3: End of file
Second file content
More data in file 2
```

**Figure 7: Test Case 5 - Concatenating multiple files**

```
amit07@DESKTOP-AMIT07:~/ppwc$ ./cat "<" "test1.txt"
Line 1: Hello World
Line 2: This is a test file
Line 3: End of file
```

**Figure 8: Test Case 6 - Input redirection from file**

```
amit07@DESKTOP-AMIT07:~/ppwc$ ./cat -n test1.txt
1 Line 1: Hello World
2 Line 2: This is a test file
3 Line 3: End of file
```

**Figure 9: Test Case 7-Line numbering option**